

DIFFERENCE-AWARE RETRIEVAL POLICIES FOR IMITATION LEARNING

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ABSTRACT

Behavior cloning offers a simple and widely used reduction of imitation learning to supervised prediction, but closed-loop deployment can expose the resulting policies to compounding errors and states outside the empirical demonstration distribution. We show that this fragility can be reduced within the original behavior-cloning data regime by retaining the demonstration corpus as a test-time retrieval memory. We introduce **Difference-Aware Retrieval Policies for Imitation Learning (DARP)**, a semi-parametric policy class that conditions action prediction on local expert neighborhoods and on the relative displacement between a query state and each retrieved neighbor. For a query state, DARP retrieves k nearest expert state-action examples, maps each tuple of neighbor state, neighbor action, and query-neighbor difference vector to a candidate action, and aggregates these candidates through a permutation-invariant operator. This design preserves the supervision assumptions of standard behavior cloning: it uses no online interaction, corrective expert feedback, or task-specific reward information. Theoretically, the retrieval-and-aggregation mechanism provides an implicit form of local Laplacian smoothing under stated manifold assumptions. Empirically, across continuous-control, simulated and real robotic-manipulation, visual-input, and multimodal-action settings, DARP improves over behavior cloning by 15-46% in the reported evaluations. These results identify difference-aware retrieval as a practical inductive bias for stabilizing imitation policies using the demonstrations already available at training time. Code and demos will be released upon acceptance.

1 INTRODUCTION

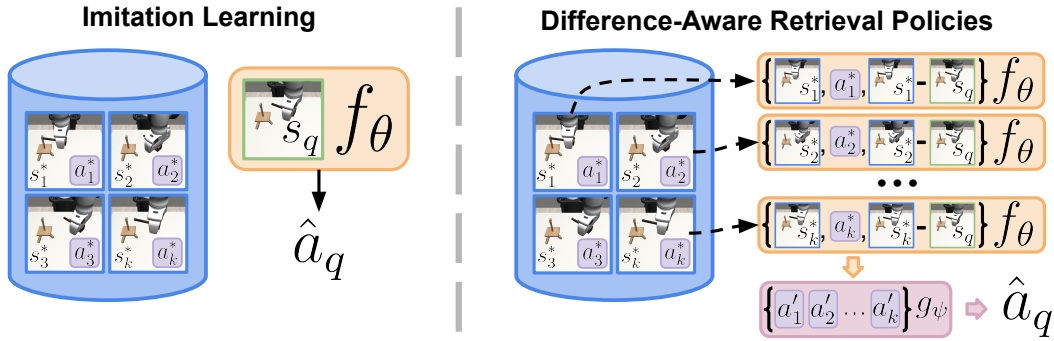


Figure 1: Overview of DARP: Standard BC (left) compresses the expert dataset into a single parametric map and discards the demonstrations at inference time. DARP (right) instead keeps the expert state-action corpus as a retrieval memory, identifies a local neighborhood around each query point s_q , and predicts actions from difference vectors between the query and retrieved neighbors.

Behavior cloning (BC) (Pomerleau, 1991) is a foundational and practically important approach to imitation learning because it turns expert demonstrations into a supervised policy-learning problem. This reduction has enabled increasingly capable robotic policies across manipulation and control domains (Zhao et al., 2023; Chi et al., 2024; Black et al., 2024; Chung et al., 2014), yet it also

exposes a persistent closed-loop weakness. During deployment, small action errors can accumulate over long-horizon execution (Ross et al., 2011) and shift the agent away from the expert-state distribution (Spencer et al., 2021; Ross et al., 2011). Once the policy enters such poorly covered regions, a purely parametric predictor may produce unstable or high-variance actions, thereby amplifying the original deviation and increasing the likelihood of task failure.

A large body of work addresses compounding error by enlarging the learner’s information set (Ross et al., 2011; Venkatraman et al., 2015; Ke et al., 2024b; Levine et al., 2020). Such methods are effective, but they often move beyond the standard BC protocol by assuming access to simulators, interactive experts, corrective labels, substantial sub-optimal data, or task-specific structure. We focus on the stricter and widely used demonstration-only regime: the learner receives expert state–action pairs and no additional online supervision. This regime motivates the central question of the paper: *how much of BC’s closed-loop fragility can be reduced by using the original demonstration dataset more effectively?*

From a statistical perspective, standard BC constrains the policy only through supervised risk on observed expert states. This pointwise objective can fit the demonstration distribution while leaving estimator behavior underconstrained in nearby low-density regions, precisely the regions that closed-loop rollouts may encounter after small deviations. A natural remedy is *smoothness*: nearby states should induce compatible action predictions unless the data indicate a behavioral discontinuity (Kobayashi, 2022; Asadi et al., 2018; Ke et al., 2024a; Chen et al., 2024). Smoothness can suppress spurious local oscillations and support rollout stability, but the mechanism used to impose it matters. We review related smoothness-based approaches in Section 4.

These mechanisms are valuable, but each leaves a gap in the pure BC setting. Data augmentation broadens coverage without necessarily enforcing local consistency; global priors can blur behaviorally distinct modes; temporal penalties regularize along trajectories rather than across nearby states; and explicit graph regularizers introduce additional smoothness hyperparameters that must be selected and tuned.

A complementary distinction is between global and local imitation. “Global” supervised policies (Black et al., 2024; Zhao et al., 2023; Chi et al., 2024) compress the demonstration corpus into a single parametric map, which is efficient but can be brittle when deployment states depart from the training manifold. “Local” methods (Pari et al., 2022; Mansimov & Cho, 2018; Salzberg & Aha, 1994) retain the dataset at inference time, retrieve similar examples, and predict actions through non-parametric or semi-parametric operations over observed expert behavior. Locality anchors decisions in demonstrations, but naive local prediction has clear limitations: it is sensitive to the retrieval metric, direct averaging can merge incompatible or multimodal actions, and absolute-state neighbor representations do not explicitly encode how the query differs from each retrieved example.

We introduce **Difference-Aware Retrieval Policies for Imitation Learning (DARP)**, a semi-parametric architecture that combines the data anchoring of local retrieval with the regularizing effect of neighborhood smoothing. At inference time, DARP (Fig. 1) retrieves k expert neighbors from the training corpus and predicts k query-specific candidate actions from tuples containing the neighbor state, the associated expert action, and the difference between the neighbor and query states. A permutation-invariant aggregation operator maps these neighbor-conditioned candidates to the executed action. This construction is deliberately conservative in its data assumptions: it uses the same demonstrations as BC and requires no online interaction, additional supervision, or task-specific knowledge. Under the assumptions stated in our analysis, the aggregation step acts as a Laplacian low-pass filter on the k -NN (k -nearest neighbor) graph of expert states, providing a parameter-free smoothing mechanism tied to the local density and geometry of the demonstration set.

The paper makes three connected contributions. First, it formulates difference-aware retrieval as a policy parameterization for imitation learning, distinguishing DARP from nearest-neighbor execution, local weighted regression, and retrieval-conditioned sequence models by explicitly learning from relative query-neighbor displacements before aggregation. Second, it relates this architecture to Laplacian regularization, showing that neighborhood aggregation implements a fixed low-pass spectral filter under stated graph-manifold assumptions. Third, it evaluates the resulting policy across continuous control (MuJoCo), simulated manipulation (Robosuite and Robocasa), visual imitation, multimodal action modeling, and a small real-robot manipulation study. Across these settings, DARP improves over behavior cloning by 15–46% in the reported comparisons, while ablations identify the

difference-vector representation and permutation-invariant aggregation as the central contributors. Taken together, the analysis and experiments support difference-aware retrieval as a practical design principle for strengthening BC without changing its data or supervision regime.

2 DIFFERENCE-AWARE RETRIEVAL POLICIES FOR IMITATION LEARNING

DARP defines a semi-parametric class of imitation-learning policies that preserves the expressive capacity of learned action predictors while grounding each decision in a retrieved local neighborhood of expert behavior. The architecture retains the standard imitation objective and the data assumptions of behavior cloning, but changes the policy parameterization so that local differences and neighborhood aggregation become part of inference. This section develops the theoretical motivation for that design; readers interested primarily in the implemented algorithm may proceed to Section 2.5 for a compact description of DARP and to Section 3 for the empirical evaluation.

We first formalize the imitation-learning setting (Section 2.1) and introduce an explicit neighbor-regularized objective (Section 2.2) that makes the role of expert-dataset geometry precise. Section 2.3 then shows how an analogous smoothing effect can be induced through architecture rather than by replacing the BC objective. Finally, Section 2.4 instantiates this principle as DARP, a practical retrieval architecture compatible with neural predictors and generative action models.

2.1 PRELIMINARIES: BEHAVIOR CLONING FOR IMITATION LEARNING

We study the standard imitation-learning problem in a finite-horizon Markov Decision Process (MDP), $\mathcal{M} = \{\mathcal{S}, \mathcal{A}, P_0\}$, where $\mathcal{S}$ is the state space, $\mathcal{A}$ is the action space, and P_0 is the initial-state distribution. A *policy* maps states to action distributions, $f_\theta : \mathcal{S} \rightarrow \Delta_{\mathcal{A}}$; when referring to the function class independently of a parameterization, we omit the subscript. The learner observes an expert demonstration dataset $\mathcal{D}^* = \{(s_j^*, a_j^*)\}$ of state-action pairs, and we use s^* and a^* for states and actions drawn from this corpus. Behavior cloning (Pomerleau, 1991) estimates f_θ through supervised maximum likelihood, $\arg \max_\theta \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\log(f_\theta(a^* | s^*))]$. Although modern policies can represent rich generative action distributions (Lipman et al., 2023; Chi et al., 2024), we begin with a Gaussian parameterization to make the analytic role of smoothing explicit¹.

2.2 WARM-UP: NEIGHBOR MANIFOLD REGULARIZED IMITATION LEARNING

Behavior cloning minimizes supervised imitation loss on states sampled from $\mathcal{D}^*$, but this pointwise objective provides no direct constraint on policy variation away from the observed expert-state manifold. During deployment, accumulated errors may therefore move the agent into nearby but underrepresented regions where an overparameterized BC policy can produce poorly controlled actions (Ross & Bagnell, 2010).

The relevant structure is the relationship among neighboring expert states and their associated actions. To make this structure explicit, we first introduce a regularized objective that encourages *local consistency*: predictions at nearby expert states should vary smoothly when the expert data support such compatibility. The resulting neighborhood-regularized loss $\mathcal{L}_{\text{MRIL}}$ augments the standard imitation objective $\mathcal{L}_{\text{BC}}$ with a smoothness term $\mathcal{L}_S$, thereby requiring the learned policy to respect the geometry of the demonstration dataset rather than relying solely on independent pointwise labels.

$$\mathcal{L}_{\text{MRIL}}(f) = \underbrace{\mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\ell(f(s^*), a^*)]}_{\text{supervised risk}(\mathcal{L}_{\text{BC}})} + \underbrace{\lambda \mathbb{E}_{s^* \sim \mathcal{D}^*} \left[\sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) \|f(s^*) - f(s_i^*)\|_2^2 \right]}_{\text{smoothness regularizer}(\mathcal{L}_S)}, \quad (1)$$

Here $\ell(f(s^*), a^*)$ denotes the supervised imitation loss, $\mathcal{N}_k(s^*)$ is the set of k nearest expert neighbors of s^* , and $w_i(s^*)$ are normalized kernel weights determined by state differences, $w_i(s^*) \propto K_\Delta\left(\frac{\|s_i^* - s^*\|}{h}\right)$. As summarized below and proved in Appendix A.1.1, this objective is a form of *manifold regularization*, equivalently Laplacian smoothing, because it penalizes high-frequency variation

¹We show that this can be relaxed in Section 2.4

over neighborhoods in the expert-state graph. Under the theorem’s smoothness and graph-limit assumptions, this regularization yields improved variance, stability, and generalization relative to unregularized BC.

Theorem 1 (Manifold Regularized BC ($\mathcal{L}_{\text{MRIL}}$) improves over vanilla BC ($\mathcal{L}_{\text{BC}}$)). *Let $f^* : \mathcal{S} \rightarrow \mathcal{A}$ be the true, underlying expert policy, assumed to be C^2 -smooth on a compact state space $\mathcal{S}$. Let $f : \mathcal{S} \rightarrow \mathcal{A}$ denote the learned policy estimator. Consider two estimators trained on expert demonstrations:*

1. **Vanilla BC:** a global supervised model minimizing

$$\mathcal{L}_{\text{BC}}(f) = \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\ell(f(s^*), a^*)].$$

2. **MRIL:** a neighbor-based estimator minimizing

$$\mathcal{L}_{\text{MRIL}}(f) = \mathcal{L}_{\text{BC}}(f) + \lambda \mathbb{E}_{s^* \sim \mathcal{D}^*} \left[\sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) \|f(s^*) - f(s_i^*)\|_2^2 \right],$$

where $w_i(s^*)$ are the kernel weights defined above and $\lambda > 0$.

Under the smoothness assumption on f , the following properties hold:

- (i) Variance reduction: The Laplacian penalty in MRIL acts as a data-dependent Tikhonov regularizer, yielding smaller estimator variance than vanilla BC.
- (ii) Smoothness guarantee: Minimizers of $\mathcal{L}_{\text{MRIL}}$ satisfy a uniform bound on the local Lipschitz constant of f , whereas vanilla BC admits interpolants with arbitrarily large Lipschitz constants between training states.
- (iii) Policy stability: In a closed-loop rollout, the deviation recursion

$$\Delta_{t+1} \leq L_s \Delta_t + L_a \|f(s_t) - f^*(s_t^*)\|^2$$

accumulates error linearly for vanilla BC, but sublinearly for MRIL, since the smoothness regularizer enforces $\|f(s^*) - f(s'^*)\| = O(\|s^* - s'^*\|)$ for neighbors s^*, s'^* .

Thus, within the theorem’s assumptions, MRIL provides stronger generalization and stability guarantees than vanilla BC.

Proof sketch. Appendix A.1.1 contains the full derivation; we state the proof strategy here. The smoothness term can be written as a graph Laplacian penalty on the k -nearest neighbor (k -NN) graph whose affinities are determined by the kernel weights w_i . In the large-sample limit, this graph penalty converges to a weighted Dirichlet energy (Belkin & Niyogi, 2008; Zhou et al., 2003). Controlling that energy enforces local Lipschitz behavior almost everywhere on the data manifold and supplies the stability argument, while the same quadratic penalty functions as Tikhonov regularization that reduces estimator variance with controlled bias under the stated assumptions. $\square$

The regularizer therefore does more than penalize isolated pairwise disagreements. It biases the policy toward smooth variation along the empirical data manifold, especially in high-density directions of the state distribution $p(s)$. In those regions, the local Lipschitz behavior of f is constrained so that small state perturbations induce controlled action changes. This provides the technical motivation for improving both in-distribution (ID) generalization and behavior on nearby out-of-distribution (OOD) states encountered during rollout.

2.3 IMPLICIT MANIFOLD REGULARIZATION VIA IN-CONTEXT ARCHITECTURES

Although MRIL clarifies the value of local smoothing, it introduces practical costs. The smoothness weight λ must be selected to trade off supervised accuracy and regularity, and the modified objective changes the optimization problem relative to standard BC. These costs motivate the architectural

² L_s and L_a are the Lipschitz constants of the environment transition dynamics with respect to state and action, respectively.

question central to DARP: *can the benefits of MRIL (Eq. 1) be obtained by modifying the policy architecture rather than the training objective?*

We answer this question with a retrieval-based architecture that induces *implicit* manifold regularization (iMRIL) while retaining the standard imitation objective. As illustrated in Fig. 2, iMRIL builds the smoothing operation into prediction itself and therefore avoids introducing λ as an additional regularization hyperparameter. Section 2.4 then turns this principle into the practical DARP algorithm.

iMRIL architecture: iMRIL transfers neighborhood aggregation from the loss into the policy parameterization. Instead of training a feed-forward predictor $f(s)$ with the explicit regularizer in Eq. 1, the architecture embeds averaging in the action predictor $\hat{f}$ and keeps the training objective in the form of supervised imitation rather than the modified objective in Eq. 1. Concretely, iMRIL learns a per-state predictor f_θ such that the neighborhood-aggregated predictor $\hat{f}(s^*) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s^*)} f_\theta(s_i^*)$, evaluated on nearest-neighbor training states $\{s_i^*\}_{i \in \mathcal{N}_k(s^*)}$, predicts the corresponding expert action a^* . With this parameterization, iMRIL optimizes:

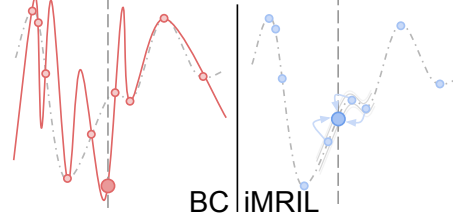


Figure 2: **iMRIL implements neighborhood smoothing through the architecture.** Under the graph interpretation, aggregation acts as an implicit Laplacian smoother that reduces high-frequency variation and encourages local consistency, whereas standard BC imposes no analogous smoothness constraint.

$$\arg \min_{\theta} \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} \left[\left\| \underbrace{\left(\frac{1}{k} \sum_{i \in \mathcal{N}_k(s^*)} f_\theta(s_i^*) \right)}_{\hat{f}(s^*)} - a^* \right\|_2 \right] \quad (2)$$

At deployment time, a query state s_q is evaluated by retrieving its k -NN from the training set and applying the same aggregation rule, $\hat{a} = \hat{f}(s_q) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} f_\theta(s_i^*)$. Section 2.4 shows that the representation used inside f is decisive in practice, motivating the difference-aware parameterization of DARP.

The essential feature is that supervision is applied after aggregation. Since $\hat{f}$ averages predictions over retrieved training states, optimizing the post-aggregation predictor discourages arbitrary disagreement among neighborhood predictions and induces the smoothing behavior described in Section 2.2. We formalize this connection through an equivalence between iMRIL and the Laplacian regularization introduced in Section 2.2.

Equivalence between iMRIL and MRIL: Appendix Section A.1 gives the full equivalence proof; here we state the main result and summarize the spectral intuition.

Theorem 2 (iMRIL is parameter-free Laplacian regularization for BC (MRIL)). *Consider the symmetric normalized k -NN graph Laplacian L (defined in Section 2.2), with eigenpairs $\{(\mu_j, u_j)\}_{j=1}^n$, where $0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_n \leq 2$.*

The minimizers of the explicit MRIL objective (Section 2.2) and the implicit iMRIL objective (Section 2.3) have the following closed form expansions

$$f_{\text{MRIL}} = \sum_{j=1}^n \frac{1}{1 + \lambda \mu_j} \langle a^*, u_j \rangle u_j \quad \hat{f}_{\text{iMRIL}} = \sum_{j=1}^n (1 - \mu_j) \langle f, u_j \rangle u_j$$

iMRIL's neighbor aggregation step applies the fixed spectral filter $\phi_{\text{iMRIL}}(\mu) = 1 - \mu$ to the graph Laplacian L , preserving low-frequency modes and suppressing high-frequency modes. The congruence between $\hat{f}_{\text{iMRIL}}$ and f_{MRIL} shows that iMRIL is equivalent to a built-in form of Laplacian smoothing (MRIL) with effective $\lambda \approx 1$ in normalized units. Unlike explicit regularization, this implicit filter requires no additional hyperparameter tuning.

Proof sketch. Appendix Section A.1.2 gives the full derivation. Diagonalizing the k -NN Laplacian yields the spectral solution of the explicit regularizer, with filter $(1 + \lambda\mu)^{-1}$ on each eigenmode. The implicit objective can be written as neighbor aggregation $\hat{f} = Sf$ with $S = D^{-1}A$, the random-walk matrix, which shares the same eigenvectors and applies the fixed filter $1 - \mu$. Both operators therefore behave as graph low-pass filters: small-eigenvalue modes, corresponding to smooth variation along the data manifold, are largely preserved, whereas large-eigenvalue modes, corresponding to rapid high-variance fluctuations between neighbors, are attenuated. Under this graph model, iMRIL performs Laplacian smoothing, reduces variance, and enforces local consistency without requiring tuning of λ . $\square$

The implicit Laplacian view does not replace policy learning; it constrains the function expressed after aggregation. The neighbor-conditioned network f_θ learns how expert actions vary under local perturbations and proposes locally adapted candidate actions for each retrieved neighbor. The aggregation operator then smooths these proposals across the neighborhood, reducing variance. In this decomposition, the learned predictor supplies local accuracy by correcting bias, while retrieval aggregation supplies stability by controlling high-frequency variation.

2.4 DIFFERENCE-AWARE RETRIEVAL POLICIES: A PRACTICAL INSTANTIATION OF iMRIL FOR IMITATION LEARNING

We now instantiate iMRIL as a practical algorithm for imitation learning. The abstract aggregation principle in Eq. 2 becomes effective through two concrete architectural choices: a difference-based parameterization of the neighbor predictor and a permutation-invariant aggregation rule for the resulting action candidates.

2.4.1 DIFFERENCE-BASED PARAMETERIZATION OF f_θ

Eq. 2 specifies the aggregation principle but leaves the representation of f_θ unspecified. The central modeling requirement is to capture how expert actions should change under local perturbations of state. Accordingly, f_θ conditions not only on a retrieved neighbor but also on the *difference* between that neighbor and the query state. In **Difference-Aware Retrieval Policies (DARP)**, the predictor receives the neighbor state s_i^* , its expert action a_i^* , and the relative vector $\Delta s_i = s_i^* - s_q$; it then outputs a query-specific action candidate $a'_i = f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q)$ for the pair consisting of query state s_q and neighbor (s_i^*, a_i^*) .

Let $\mathcal{N}_k(s_q)$ denote the index set of the k nearest neighbors under a distance function $d(s_q, s_i^*)$.³ Given this retrieved neighborhood, DARP applies the aggregation principle from Section 2.3 to predict an action for s_q :

$$\hat{a}_q = f_{\text{DARP}}(s_q) = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} a'_i \quad (3)$$

$$= \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q). \quad (4)$$

At training time, this architecture yields the following imitation objective over the expert dataset $\mathcal{D}^*$:

$$\arg \min_{\theta} \mathbb{E}_{(s_q^*, a_q^*) \sim \mathcal{D}^*} \left[\|f_{\text{DARP}}(s_q^*) - a_q^*\|^2 \right] \quad (5)$$

Optimization updates f_θ by minimizing the discrepancy between the aggregated prediction $\hat{a}_q$ and the expert action a_q^* . Because the loss remains the standard supervised imitation objective, f_θ can be implemented with conventional feedforward or convolutional neural networks. Section 3 shows empirically that the difference-based input representation in Eq. 3 is central to performance. At inference time, DARP retrieves the k -NN of the current state and executes the action obtained by the aggregation operation in Eq. 3.

³In our work we use the Euclidean distance in a pre-trained embedding space, although other neighborhood functions are also applicable. We refer the reader to Appendix Section A.2.1 for a thorough discussion of design decisions in constructing neighborhood sets via retrieval.

2.4.2 GOING BEYOND LINEAR AGGREGATION

The preceding architecture uses the simple average $\hat{a}_q = \frac{1}{k} \sum_{i \in \mathcal{N}_k(s_q)} a'_i$, but averaging is one instance of a broader class of permutation-invariant aggregation functions $g_\psi(\{a'_i\}_{i \in \mathcal{N}_k(s_q)})$. Set-compliant architectures such as set transformers (Lee et al., 2019) or DeepSets (Zaheer et al., 2017) can parameterize g_ψ and yield the generalized predictor $\hat{a}_q = g_\psi(\{f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q)\}_{i \in \mathcal{N}_k(s_q)})$. Beyond additional expressivity, a parametric aggregator allows DARP to represent richer action distributions, such as Gaussian mixture models (Pignat & Calinon, 2019) or diffusion models (Chi et al., 2024), rather than the unimodal Gaussian structure associated with the L_2 objective in Eq. 5. In this formulation, DARP predicts distribution parameters α for $p(a_q; \alpha)$, including mixture means, covariances, and weights, or a diffusion-model score function, and can therefore be trained by maximum likelihood on multimodal action distributions:

$$\arg \max_{\theta} \mathbb{E}_{(s_q^*, a_q^*) \sim \mathcal{D}^*} [\log p(a_q^*; \alpha_\theta(s_q^*))], \text{ where } \alpha_\theta(s_q^*) = g_\psi(\{f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q^*)\}_{i \in \mathcal{N}_k(s_q^*)}) \quad (6)$$

For a query state s_q , inference samples $\hat{a}_q \sim p(\cdot; \alpha_\theta(s_q))$, where $\alpha_\theta(s_q) = g_\psi(\{f_\theta(s_i^*, a_i^*, \Delta s_i = s_i^* - s_q)\}_{i \in \mathcal{N}_k(s_q)})$ is computed from the neighbors retrieved at test time. Appendix Section A.3 gives the corresponding training pseudocode.

2.5 SUMMARY

The preceding development provides the technical basis for DARP. Local consistency in BC can be expressed as manifold regularization, or Laplacian smoothing, which improves variance, smoothness, and stability relative to standard BC under the assumptions of Theorem 1. Instead of adding an explicit penalty and tuning λ , DARP embeds neighborhood aggregation into the policy architecture, inducing the corresponding smoothing effect implicitly (Theorem 2). The resulting algorithm is simple: for each expert pair (s_q^*, a_q^*) , DARP retrieves the k nearest neighbor states from $\mathcal{D}^*$, computes difference vectors $\Delta s_i = s_i^* - s_q^*$, maps each tuple $(s_i^*, a_i^*, \Delta s_i)$ through f_θ to produce candidate actions, and aggregates those candidates with a permutation-invariant function g_ψ to obtain the final prediction. Training remains standard supervised imitation learning.

3 EXPERIMENTAL EVALUATION

We evaluate DARP through three questions that align the architectural motivation with measurable empirical behavior: **Q1**: Does DARP improve over standard behavior cloning across the evaluated domains?, **Q2**: Does the same difference-aware retrieval mechanism remain effective with more complex state representations and action distributions?, and **Q3**: Which architectural components are responsible for the observed gains? The evaluation spans low-dimensional state inputs, visual feature representations, and diverse action models across MuJoCo continuous control, Robosuite and RoboCasa manipulation, a small real-world manipulation study, and discontinuous environments designed to stress-test neighbor-based prediction.

3.1 BASELINE COMPARISONS AND TASK DESCRIPTIONS

MuJoCo Tasks: The MuJoCo (Todorov et al., 2012; Fu et al., 2020) suite evaluates locomotion control for articulated agents on a flat plane: Hopper (single-legged hopping robot), Walker (bipedal humanoid), Ant (quadruped), and HalfCheetah (biped).

Robosuite Tasks: The Robosuite (Zhu et al., 2020) tasks evaluate single-arm object manipulation. Stack requires placing a smaller cube on a larger cube; Threading requires inserting a thin, needle-like tool into a small ring; and Square Peg requires placing a square wooden block with a central hole onto a square peg.

RoboCasa Tasks: The RoboCasa (Nasiriany et al., 2024) tasks evaluate single-arm manipulation in randomized kitchen scenes. Drawer requires closing an open drawer, Door requires closing an open microwave door, and Stove requires turning a knob to switch off a stove burner.

Baseline Comparisons: We compare DARP with retrieval, parametric, and regularized alternatives that isolate different uses of the demonstration corpus: (1) **R&P** (Sridhar et al., 2025): directly executes the action associated with the nearest neighbor; (2) **LWR** (Pari et al., 2022): performs locally weighted regression over retrieved neighbors; (3) **BC**: denotes standard parametric behavior cloning; (4) **REGENT** (Sridhar et al., 2025): is a transformer-based in-context method conditioned on retrieved neighbors; and (5) **MRIL**: is the explicitly smoothed objective introduced in Section 2.2.

3.2 CAN DARP CONSISTENTLY OUTPERFORM STANDARD BEHAVIOR CLONING? (Q1)

We first test the main empirical claim in low-dimensional state spaces, where state distances are directly defined and retrieval neighborhoods are readily interpretable. The evaluation includes MuJoCo locomotion and Robosuite/RoboCasa manipulation tasks, with manipulation demonstrations generated using MimicGen (Mandlekar et al., 2023). Unless otherwise specified, the aggregation function g is the simple average of all neighbor-conditioned action predictions a' .

Method	Hopper	Ant	Walker	HalfCheetah
R&P (Sridhar et al., 2025)	711.82 $\pm$ 85.63	-305.97 $\pm$ 76.42	419.18 $\pm$ 50.21	-178.64 $\pm$ 29.75
LWR (Pari et al., 2022)	1703.78 $\pm$ 245.95	846.59 $\pm$ 216.06	1484.91 $\pm$ 356.54	1945.82 $\pm$ 567.26
BC	2313.65 $\pm$ 203.75	2376.20 $\pm$ 339.43	2658.40 $\pm$ 274.08	1063.23 $\pm$ 371.08
REGENT (Sridhar et al., 2025)	1819.39 $\pm$ 186.24	-302.10 $\pm$ 146.67	507.01 $\pm$ 76.10	169.85 $\pm$ 63.10
MRIL	2793.63 $\pm$ 156.41	3869.08 $\pm$ 241.00	4370.96 $\pm$ 168.13	701.58 $\pm$ 195.08
DARP	3545.57 $\pm$ 3.54	4383.28 $\pm$ 266.37	4894.01 $\pm$ 75.12	5515.41 $\pm$ 841.33
DARP Set Transformer	2965.86 $\pm$ 103.08	4063.79 $\pm$ 218.80	4752.42 $\pm$ 109.23	3417.85 $\pm$ 764.57

Table 1: DARP is the best-performing MuJoCo method in the reported comparison, with DARP Set Transformer also competitive. The table compares DARP, BC, retrieval baselines, REGENT, and MRIL on low-dimensional MuJoCo state inputs. Scores are averaged over 100 independent trials with 95% confidence intervals. Parametric policies use Multi-Layer Perceptrons; diffusion-policy results appear in Section 8.

Across the low-dimensional evaluations, DARP improves over standard BC in every reported manipulation task and achieves the strongest MuJoCo scores. On robotic manipulation, the absolute gains over BC range from 15–25 percentage points; on locomotion, DARP yields large return increases relative to BC (Table 1 and Table 2). Purely non-parametric retrieval methods (R&P and LWR) are substantially weaker on this suite, indicating that retrieval alone does not explain the gains. MRIL often improves over vanilla BC, but the best overall results are obtained by the DARP architecture, supporting the value of embedding local smoothing directly into the policy parameterization.

Having established gains in compact state spaces, and following the practical instantiation in Section 2.4, we next evaluate whether the same difference-aware retrieval mechanism remains effective with higher-dimensional representations and richer action models.



Figure 3: The “DrawerClose” RoboCasa task, in which the robotic arm must close an open drawer.



Figure 4: The “Threading” Robosuite task, in which the robotic arm must insert a needle-like implement into a small hole.

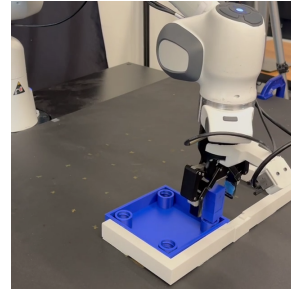


Figure 5: The real-world FurnitureBench square-table assembly task, in which the robot must pick up a table leg and screw it into a hole at the tabletop corner.

Method	RoboCasa			Robosuite			Real
	Drawer	Door	Stove	Stack	Thrd.	Peg	Sq. Table
BC	54	29	28	47	37	46	44
DARP	85	45	43	72	63	62	92

Table 2: Comparison of DARP and BC across RoboCasa, Robosuite, and the real-world square-table task using low-dimensional state features. Scores are success percentages (RoboCasa & Robosuite: 100 trials; Real: 50 trials). DARP outperforms BC on every task in this table, including increasing real-world success from 44% to 92%. RoboCasa and Robosuite policies use Multi-Layer Perceptrons, while the real-world results use diffusion policy; see Section 8.

3.3 CAN DARP HANDLE MORE COMPLEX STATE REPRESENTATION AND ACTION DISTRIBUTIONS? (Q2)

High-Dimensional Visual Input Representations. To evaluate DARP beyond compact state vectors, we train simulated manipulation policies using R3M image embeddings (Nair et al., 2022). This setting tests whether local retrieval structure remains useful when states are represented as high-dimensional visual features rather than hand-specified state coordinates (Table 3). DARP outperforms BC on all three vision-based Robosuite tasks, and its average improvement of $\sim 35\%$ is larger than the corresponding average improvement on low-dimensional Robosuite states ($\sim 22\%$). These results show that the difference-aware parameterization remains effective when retrieval operates in a learned visual feature space.

Method	Stack	Thrd.	Peg
BC	44	38	17
DARP	75	76	52

Table 3: Success rates (%) for BC and DARP on vision-based Robosuite tasks using R3M image features.

Multi-modal Action Distributions. DARP also supports multimodal action modeling. On Push-T, where the policy uses a Set Transformer head to predict Gaussian-mixture parameters, DARP performs more than 20% better than behavior cloning. Appendix A.2.2 provides the experimental details.

3.4 HOW DO DIFFERENT ARCHITECTURAL COMPONENTS CONTRIBUTE TO DARP’S PERFORMANCE GAINS? (Q3)

Ablation Study: We isolate the architectural choices responsible for DARP’s gains through a systematic ablation study. The variants are: (1) standard DARP; (2) DARP without neighbor actions; (3) an ensemble of 10 BC agents; (4) DARP with random neighbors rather than distance-based retrieval; (5) DARP using the L_2 norm of the distance vector; (6) the BC baseline using only the query state s_q ; (7) DARP using the query state rather than the difference vector between query and neighbor

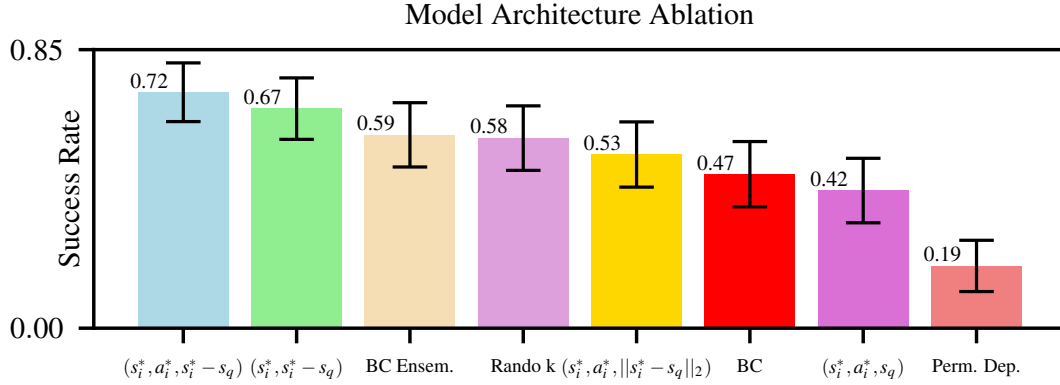


Figure 6: **Distance vectors and permutation invariance are central to DARP’s performance.** Ablating the core architecture shows that the tuple representation $(s_i^*, a_i^*, s_i^* - s_q)$ and permutation-invariant aggregation account for the strongest gains. Success rate is averaged across 100 trials on the Robosuite Stack environment with 95% confidence intervals.

states; and (8) DARP with a permutation-dependent, rather than permutation-invariant, aggregator for the a' candidates. Figure 6 reports the comparison on Robosuite Stack.

The ablations identify two primary mechanisms. First, the relative difference vector is more informative than random neighbor selection or representations that discard directional information, indicating that DARP benefits from meaningful local geometry rather than from retrieval as an undifferentiated memory lookup. Second, the permutation-invariant aggregation function g is crucial: permutation-dependent alternatives substantially reduce performance. Neighbor actions provide a smaller additional benefit, suggesting that the dominant gains arise from difference-aware local adaptation combined with order-invariant smoothing.

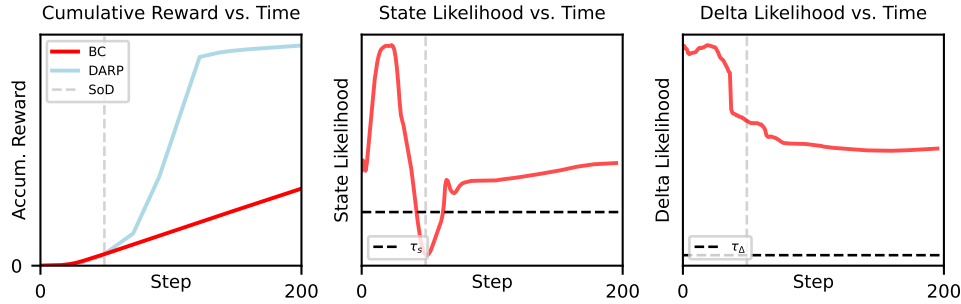


Figure 7: **Cumulative rewards for BC and DARP on the Robosuite Stack task show initially aligned rollouts that diverge when BC fails and DARP succeeds.** The vertical dashed line marks the step of divergence, labeled “SoD”. At the SoD, the state likelihood is $< \tau_s$ (OOD), while the delta likelihood is $> \tau_\Delta$ (in distribution).

Divergence Analysis: To interpret the performance gap between DARP and BC, we examine rollouts in which BC fails but DARP succeeds and define the “step of divergence” as the first time at which the two policies begin to receive substantially different reward. We define τ_s and τ_Δ as the 1st percentile of training-set likelihoods, so that 1% of training deltas have likelihood below τ_Δ . Across all six analyzed rollouts from Robosuite Stack and MuJoCo Hopper, the query state at the SoD has state likelihood $< \tau_s$ but delta likelihood $> \tau_\Delta$. This pattern supports the interpretation that DARP gains partly from a representation in which states that are slightly OOD in absolute coordinates can remain in distribution relative to retrieved neighbors. Figure 7 shows reward drift, SoDs, and state/delta likelihoods for one task, and Appendix A.2.4 reports additional robustness experiments.

4 RELATED WORK

Non-Parametric Imitation Learning Methods: Non-parametric IL methods establish that local structure in demonstration data can provide a strong inductive bias for imitation. VINN Pari et al. (2022) applies locally weighted regression and obtains strong results in image-embedding spaces,

while MiDiGaP von Hartz et al. (2025) uses mixtures of Gaussian processes to represent multimodal trajectories and support rapid generalization. SEABO Lyu et al. (2024) incorporates retrieval into offline RL by rewarding transitions close to retrieved neighbors. FlowRetrieval Lin et al. (2024), STRAP Memmel et al. (2025), and Behavior Retrieval Du et al. (2023) combine non-parametric retrieval with finetuning from large unlabeled datasets, enabling generalization through test-time training. DARP is closest in spirit to these retrieval-based approaches but differs in its learned semi-parametric prediction rule: it operates on retrieved tuples $(s_i^*, a_i^*, s_i^* - s_q)$ and aggregates query-conditioned candidate actions rather than relying on direct non-parametric averaging or test-time adaptation. This design preserves the anchoring benefit of local prediction while retaining the flexibility of learned parametric policies.

Smoothness-Constrained Policy Learning: Recent work uses explicit smoothness constraints to improve policy stability and robustness. L2C2 Kobayashi (2022) studies model-free RL under local Lipschitz-continuity constraints, improving smoothness and noise robustness without sacrificing expressiveness, and Asadi et al. (2018) develop related Lipschitz constraints for model-based RL. CCIL Ke et al. (2024a) extends this principle to imitation learning by generating synthetic corrective labels with a Lipschitz-constrained dynamics model. Related smoothness mechanisms have also been scaled to humanoid controllers Chen et al. (2024) to reduce deployment shakiness. DARP addresses the same stability motivation through a different mechanism: it induces local smoothing through retrieval-conditioned aggregation while retaining the standard imitation-learning objective and avoiding an explicit smoothness penalty.

In-Context Learning Methods: A related body of work treats retrieval as in-context imitation learning. REGENT Sridhar et al. (2025) studies retrieval-augmented generalization by conditioning a causal transformer on retrieved states, actions, and rewards, whereas DPT Lee et al. (2023) uses supervised transformer pretraining to predict actions from query states and in-context datasets, effectively learning exploratory behavior. Other in-context architectures include ICRT Fu et al. (2024), Instant Policy Vosylius & Johns (2025), and KAT Di Palo & Johns (2024). These methods primarily emphasize rapid adaptation to new tasks and environments. DARP instead focuses on improving stability and performance within the standard imitation-learning setting by combining retrieval with explicit query-neighbor difference modeling and permutation-invariant aggregation.

5 CONCLUSION

We introduced **Difference-Aware Retrieval Policies for Imitation Learning (DARP)** (DARP), a semi-parametric policy architecture that reparameterizes imitation learning around relative differences between query states and retrieved expert neighbors rather than a direct global state-to-action map. The analysis connects this architecture to implicit Laplacian smoothing, providing a principled account of how retrieval-conditioned aggregation can reduce high-frequency variation and encourage local consistency under the stated assumptions. The empirical evaluation across continuous control, simulated and real robotic manipulation, visual representations, and multimodal action settings supports three conclusions. First, DARP improves over standard BC in the reported low-dimensional state evaluations. Second, the same design remains effective across the tested representation and action-model settings, with improvements of 15-46% in the evaluated scenarios. Third, ablations identify relative difference vectors and permutation-invariant aggregation as the key components behind these gains. Overall, the results establish difference-aware retrieval as a practical and data-conservative mechanism for strengthening imitation policies while preserving the assumptions and training objective of behavior cloning.

6 REPRODUCIBILITY STATEMENT

Upon acceptance, we will release the source code, policy and environment configuration files, MuJoCo experiment data, scripts for generating Robosuite expert demonstrations with MimicGen, and modality-transformation code used in the experiments. On NVIDIA L40 and L40s GPUs, these materials reproduce the reported results, with the exception of transformer-based runs (REGENT and Set Transformer), for which non-determinism may produce slight deviations from the numbers reported in the paper.

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A APPENDIX

A.1 LEMMAS AND PROOFS

A.1.1 PROOF OF THEOREM 1

The proof of Theorem 1 begins by expressing the smoothness term as a standard graph-Laplacian penalty (Chung, 1997; Zhou et al., 2003; Belkin & Niyogi, 2008).

Lemma 1 (Smoothness regularizer as k -NN graph Laplacian penalty). *Let $\{s_1^*, \dots, s_n^*\}$ be the expert states with corresponding predicted actions $f(s_i^*) \in \mathbb{R}^{d_a}$. For each i , let $\mathcal{N}_k(s_i^*)$ denote the indices of the k -nearest neighbors of s_i^* (excluding i). Define asymmetric weights*

$$\tilde{W}_{ij} = \begin{cases} w_j(s_i^*), & \text{if } j \in \mathcal{N}_k(s_i^*), \\ 0, & \text{otherwise,} \end{cases}$$

and construct a symmetric affinity matrix

$$W_{ij} = \frac{1}{2} (\tilde{W}_{ij} + \tilde{W}_{ji}).$$

Let D be the degree matrix with $D_{ii} = \sum_j W_{ij}$, and define the k -NN graph Laplacian $L = D - W$.

Then the smoothness regularizer can be written as the quadratic form

$$\mathcal{L}_S(f) = \frac{1}{n} \sum_{i=1}^n \sum_{j \in \mathcal{N}_k(s_i^*)} w_j(s_i^*) \|f(s_i^*) - f(s_j^*)\|^2 \propto \text{Tr}(F^\top L F),$$

where $F = [f(s_1^*), f(s_2^*), \dots, f(s_n^*)]^\top \in \mathbb{R}^{n \times d_a}$. Equivalently, in the scalar case,

$$\mathcal{L}_S(f) \propto f^\top L f.$$

Corollary 1 (Continuum limit of smoothness regularizer). *Assume states $\{s_i^*\}_{i=1}^n$ are sampled i.i.d. from a smooth density $p(s^*)$ supported on an m -dimensional C^2 manifold $\mathcal{M} \subset \mathbb{R}^d$. Let W be the symmetrized k -NN affinity matrix constructed from a kernel K_Δ with bandwidth h , and let $L = D - W$ be the graph Laplacian.*

If $n \rightarrow \infty$, $h \rightarrow 0$, and $nh^{m+2} \rightarrow \infty$, then the normalized quadratic form converges to the weighted Dirichlet energy:

$$\frac{1}{n^2 h^{m+2}} \text{Tr}(F^\top L F) \longrightarrow C_K \int_{\mathcal{M}} \|\nabla_{\mathcal{M}} f(s)\|_2^2 p(s)^2 d\text{vol}(s),$$

where $C_K > 0$ is a constant depending only on the kernel K_Δ .

Theorem 1 (Manifold Regularized BC ($\mathcal{L}_{\text{MRIL}}$) improves over vanilla BC ($\mathcal{L}_{\text{BC}}$)). *Let $f^* : \mathcal{S} \rightarrow \mathcal{A}$ be the true, underlying expert policy, assumed to be C^2 -smooth on a compact state space $\mathcal{S}$. Let $f : \mathcal{S} \rightarrow \mathcal{A}$ denote the learned policy estimator. Consider two estimators trained on expert demonstrations:*

1. **Vanilla BC:** a global supervised model minimizing

$$\mathcal{L}_{\text{BC}}(f) = \mathbb{E}_{(s^*, a^*) \sim \mathcal{D}^*} [\ell(f(s^*), a^*)].$$

2. **MRIL:** a neighbor-based estimator minimizing

$$\mathcal{L}_{\text{MRIL}}(f) = \mathcal{L}_{\text{BC}}(f) + \lambda \mathbb{E}_{s^* \sim \mathcal{D}^*} \left[\sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) \|f(s^*) - f(s_i^*)\|_2^2 \right],$$

where $w_i(s^*)$ are the kernel weights defined above and $\lambda > 0$.

Under the smoothness assumption on f , the following properties hold:

- (i) Variance reduction: *The Laplacian penalty in MRIL acts as a data-dependent Tikhonov regularizer, yielding smaller estimator variance than vanilla BC.*

(ii) Smoothness guarantee: Minimizers of $\mathcal{L}_{\text{MRIL}}$ satisfy a uniform bound on the local Lipschitz constant of f , whereas vanilla BC admits interpolants with arbitrarily large Lipschitz constants between training states.

(iii) Policy stability: In a closed-loop rollout, the deviation recursion

$$\Delta_{t+1} \leq L_s \Delta_t + L_a \|f(s_t) - f^*(s_t^*)\|^4$$

accumulates error linearly for vanilla BC, but sublinearly for MRIL, since the smoothness regularizer enforces $\|f(s^*) - f(s'^*)\| = O(\|s^* - s'^*\|)$ for neighbors s^*, s'^* .

Thus, within the theorem's assumptions, MRIL provides stronger generalization and stability guarantees than vanilla BC.

Proof. We establish the three claims separately.

(i) Variance reduction. The Laplacian penalty in $\mathcal{L}_{\text{MRIL}}$ is

$$\sum_{i,j} W_{ij} \|f(s_i^*) - f(s_j^*)\|^2 = 2f^\top L f,$$

where $L = D - W$ is the graph Laplacian and W is the k -NN affinity matrix. By Lemma 1, this expression is the empirical Dirichlet energy of f on the k -NN graph. Standard results on manifold regularization (Zhou et al., 2003; Belkin & Niyogi, 2008) identify this quadratic penalty with Tikhonov regularization in the graph Laplacian norm $\|f\|_L^2 = f^\top L f$. Under the regularized-estimation assumptions used in the theorem, the penalty reduces estimator variance relative to the unregularized solution while controlling the induced bias. Thus MRIL has smaller estimator variance than vanilla BC, which contains no corresponding graph penalty.

(ii) Smoothness guarantee. Consider the continuum limit (Corollary 1): for i.i.d. samples $\{s_i^*\}$ from density p on a smooth manifold $\mathcal{M}$, the normalized penalty converges to

$$\int_{\mathcal{M}} \|\nabla f(s)\|^2 p(s)^2 d\text{vol}(s).$$

The limiting quantity is the weighted Dirichlet energy of f on $\mathcal{M}$. If this integral is finite, f belongs to the Sobolev space $H^1(\mathcal{M}, p^2)$ and is locally Lipschitz almost everywhere with

$$\|f(s^*) - f(s'^*)\| \leq C \|s^* - s'^*\| \quad \text{for } p\text{-a.e. neighbor pairs } s^*, s'^*.$$

Consequently, minimizers of $\mathcal{L}_{\text{MRIL}}$ have bounded local Lipschitz behavior along high-density regions of the state space. Vanilla BC imposes no analogous constraint: an oscillatory interpolant can match the training data exactly and obtain the same supervised risk while having an arbitrarily large Lipschitz constant between observed states.

(iii) Policy stability. Let $\Delta_t = \|s_t - s_t^*\|$ denote the deviation at time t . For Lipschitz dynamics T ,

$$\Delta_{t+1} \leq L_s \Delta_t + L_a \|f(s_t) - f^*(s_t^*)\|.$$

Decompose the action error:

$$\|f(s_t) - f^*(s_t^*)\| \leq \|f(s_t) - f^*(s_t)\| + \|f^*(s_t) - f^*(s_t^*)\|.$$

For vanilla BC, the first term $\|f(s_t) - f^*(s_t)\|$ is minimized only on the empirical distribution $P_{\mathcal{S}}$; off distribution, it can remain $O(1)$ independently of Δ_t . The second term satisfies $\|f^*(s_t) - f^*(s_t^*)\| = O(\Delta_t)$ by smoothness of f^* . Hence the recursion can take the form

$$\Delta_{t+1} \leq L_s \Delta_t + L_a (O(1) + O(\Delta_t)),$$

which accumulates linearly in t .

⁴ L_s and L_a are the Lipschitz constants of the environment transition dynamics with respect to state and action, respectively.

For MRIL, the Laplacian penalty enforces

$$\|f(s^*) - f(s'^*)\| \leq C\|s^* - s'^*\| \quad \text{for neighbor pairs } (s^*, s'^*),$$

as shown in part (ii). Thus $\|f(s_t) - f^*(s_t)\| = O(h^2)$ (where h is the radius of the kernel bandwidth around the point s_t) by local-linear regression error bounds, and $\|f^*(s_t) - f^*(s_t^*)\| = O(\Delta_t)$ by smoothness of f^* . Combining these,

$$\Delta_{t+1} \leq L_s \Delta_t + L_a(O(h^2) + O(\Delta_t)).$$

Under the theorem's Lipschitz and bandwidth conditions, the constant multiplying Δ_t is smaller with the smoothness constraint, so cumulative error grows more slowly than in the unregularized BC recursion. In particular, for sufficiently small h , this argument gives sublinear growth in the rollout horizon, whereas the vanilla BC bound permits linear accumulation.

Conclusion. Claims (i)–(iii) establish lower variance, local smoothness control, and improved rollout-error accumulation for MRIL relative to vanilla BC under the stated assumptions, completing the proof. $\square$

Kernel choice. For the IC smoothness regularizer, we adopt a Gaussian kernel

$$w_i(s^*) \propto \exp\left(-\frac{\|s^* - s_i^*\|^2}{2h^2}\right), \quad \sum_{i \in \mathcal{N}_k(s^*)} w_i(s^*) = 1,$$

with bandwidth h set to the median distance to the k -th nearest neighbor across the dataset. This choice is standard in manifold regularization (Belkin & Niyogi, 2008; Zhou et al., 2003) and ensures that the graph Laplacian penalty converges to the Dirichlet energy in the continuum limit. In practice, this default is stable across the evaluated tasks, and alternative kernels such as uniform k -NN weights or exponential decay yield qualitatively similar results.

A.1.2 PROOF OF THEOREM 2

We begin with two lemmas that specify the spectral form of explicit Laplacian regularization and neighbor aggregation.

Lemma 2 (Spectral form of explicit Laplacian regularization). *Let L be the symmetric normalized graph Laplacian with eigenpairs $\{(\mu_j, u_j)\}_{j=1}^n$, where $0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_n \leq 2$. The minimizer of the penalized objective*

$$\mathcal{L}_\lambda(f) = \|f - a^*\|^2 + \lambda f^\top L f$$

has the closed-form expansion

$$f_\lambda = \sum_{j=1}^n \frac{1}{1 + \lambda \mu_j} \langle a^*, u_j \rangle u_j.$$

Thus λ directly determines the spectral filter $\phi_\lambda(\mu) = (1 + \lambda \mu)^{-1}$ applied to each Laplacian mode.

Proof. Diagonalize $L = U\Lambda U^\top$ with $U = [u_1, \dots, u_n]$ orthogonal and $\Lambda = \text{diag}(\mu_1, \dots, \mu_n)$. Write $f = Uc$, $a^* = Ub$ in this basis. The objective becomes

$$\|Uc - Ub\|^2 + \lambda c^\top \Lambda c = \sum_{j=1}^n (c_j - b_j)^2 + \lambda \mu_j c_j^2.$$

Minimizing each term yields $c_j = \frac{1}{1 + \lambda \mu_j} b_j$. Transforming back gives the stated expansion. $\square$

Lemma 3 (Spectral form of neighbor aggregation). *Let $S = D^{-1}A$ be the random-walk matrix of the k -NN graph, with adjacency A and degree D . For any prediction vector f , the neighbor-averaged prediction is $\hat{f} = Sf$. In the Laplacian eigenbasis, this corresponds to the spectral filter*

$$\hat{f} = \sum_{j=1}^n (1 - \mu_j) \langle f, u_j \rangle u_j,$$

i.e. $\phi_{\text{DARP}}(\mu) = 1 - \mu$.

Proof. By definition, $L = I - D^{-1/2}AD^{-1/2}$ and $S = D^{-1}A = I - L_{\text{rw}}$ where $L_{\text{rw}} = D^{-1}L$ is the random-walk Laplacian. Since L_{rw} and L share the same spectrum up to similarity transform, the eigenbasis $\{u_j\}$ diagonalizes S . Thus for each mode u_j , $Su_j = (1 - \mu_j)u_j$, yielding the claimed spectral filter. $\square$

Theorem 2 (iMRIL is parameter-free Laplacian regularization for BC (MRIL)). *Consider the symmetric normalized k -NN graph Laplacian L (defined in Section 2.2), with eigenpairs $\{(\mu_j, u_j)\}_{j=1}^n$, where $0 = \mu_1 \leq \mu_2 \leq \dots \leq \mu_n \leq 2$.*

The minimizers of the explicit MRIL objective (Section 2.2) and the implicit iMRIL objective (Section 2.3) have the following closed form expansions

$$f_{\text{MRIL}} = \sum_{j=1}^n \frac{1}{1 + \lambda \mu_j} \langle a^*, u_j \rangle u_j \quad \hat{f}_{\text{iMRIL}} = \sum_{j=1}^n (1 - \mu_j) \langle f, u_j \rangle u_j$$

iMRIL's neighbor aggregation step applies the fixed spectral filter $\phi_{\text{iMRIL}}(\mu) = 1 - \mu$ to the graph Laplacian L , preserving low-frequency modes and suppressing high-frequency modes. The congruence between $\hat{f}_{\text{iMRIL}}$ and f_{MRIL} shows that iMRIL is equivalent to a built-in form of Laplacian smoothing (MRIL) with effective $\lambda \approx 1$ in normalized units. Unlike explicit regularization, this implicit filter requires no additional hyperparameter tuning.

Proof. From Lemma 3, the neighbor aggregation operator $S = D^{-1}A$ acts on Laplacian eigenmodes u_j as

$$Su_j = (1 - \mu_j)u_j,$$

where μ_j are the normalized Laplacian eigenvalues. Thus in the graph Fourier basis, neighbor aggregation corresponds to multiplying each mode by the fixed spectral filter $\phi_{\text{DARP}}(\mu) = 1 - \mu$.

On the other hand, Lemma 2 shows that explicit Laplacian regularization with parameter λ yields the spectral filter $\phi_\lambda(\mu) = (1 + \lambda \mu)^{-1}$. Both filters downweight high-frequency modes ($\mu \gg 0$) while preserving low-frequency modes ($\mu \approx 0$). The key difference is that $\phi_\lambda(\mu)$ requires tuning λ , whereas $\phi_{\text{DARP}}(\mu)$ is parameter-free.

To see the equivalence, note that for small μ ,

$$\phi_{\text{DARP}}(\mu) = 1 - \mu \approx (1 + \mu)^{-1} = \phi_{\lambda=1}(\mu) \quad \text{up to } O(\mu^2) \text{ terms.}$$

Thus DARP can be interpreted as performing Laplacian smoothing with an effective regularization weight of order $\lambda \approx 1$ in normalized units. Moreover, for large μ , $\phi_{\text{DARP}}(\mu)$ damps high-frequency modes even more strongly by driving them toward zero, providing a sharper low-pass effect than explicit regularization.

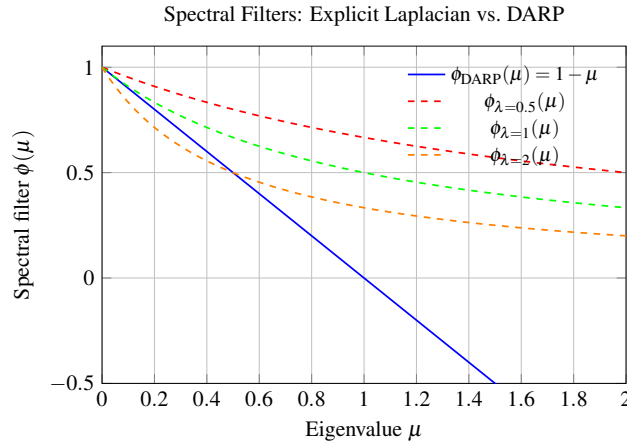


Figure 8: DARP achieves sharper low-pass filtering

Therefore, within this graph-spectral formulation, DARP’s aggregation step is equivalent to implicit Laplacian regularization with fixed spectral filter ϕ_{DARP} , eliminating the need to tune λ explicitly. $\square$

This perspective treats DARP as *locally adaptive implicit regularization*: rather than introducing a global regularization weight λ , neighborhood aggregation imposes smoothing through the graph structure itself. The effective regularization depends on local degree and neighborhood geometry, and therefore reflects the density of the expert demonstrations. Spectrally, this corresponds to the fixed filter $\phi_{\text{DARP}}(\mu) = 1 - \mu$, which more strongly attenuates high-frequency modes in this normalized formulation. Figure 8 illustrates the comparison with explicit Laplacian filters.

A.2 ADDITIONAL EXPERIMENTAL DETAILS

A.2.1 RETRIEVAL

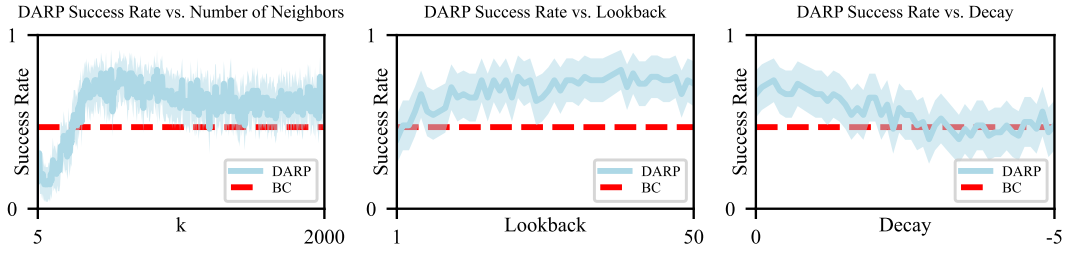


Figure 9: DARP sensitivity to retrieval hyperparameters on Robosuite Stack. Left: performance is poor with very few neighbors, peaks near 500 retrieved neighbors, and plateaus slightly above BC as k approaches the dataset size. Center: using more history generally improves performance, while little or no history can underperform BC. Right: performance depends on the exponential decay applied to older observations; excessive decay approximates a no-history retrieval rule and degrades performance. Success rates are measured over 50 trials with 95% confidence intervals.

The distance function $d(s_q, s_i^*)$ determines which expert examples enter the local neighborhood and is therefore an important implementation choice. Although Euclidean distance between current states can be effective, our experiments use a history-aware variant that exploits the sequential structure of demonstrations.

Let the query trajectory be $S_q = (\dots, s_{q,-1}, s_{q,0})$, where $s_{q,0}$ is the current query state s_q . To compute $d(s_q, s_i^*)$ for an expert state s_i^* , we identify the expert trajectory S_j^* containing that state and its time index t , so that $s_i^* = s_{j,t}^*$. Given a lookback parameter ℓ specifying the number of previous states included in retrieval, we define:

$$d(s_q, s_i^*) = \sum_{n=0}^{\ell-1} \|s_{q,-n} - s_{j,t-n}^*\|$$

This distance accumulates Euclidean discrepancies between the current and previous $\ell - 1$ states of the query trajectory and the corresponding expert trajectory, assuming valid indices. To emphasize recent observations during retrieval, we also use an exponentially decayed version with decay rate $\gamma \geq 0$:

$$d(s_q, s_i^*) = \sum_{n=0}^{\ell-1} \|s_{q,-n} - s_{j,t-n}^*\| \cdot e^{-\gamma n}$$

Figure 9 reports how success rate changes as k , lookback, and decay are varied.

A.2.2 CAN DARP HANDLE TASKS REQUIRING THE REPRESENTATION OF MULTI-MODAL ACTION DISTRIBUTIONS?

We evaluate DARP on Push-T (Chi et al., 2024), a task that requires representing multimodal action distributions (Figure 10). In this experiment, DARP uses a Set Transformer head to predict the parameters of a Gaussian Mixture Model. With this GMM head, DARP improves over BC by 22% on Push-T (Table 4), showing that the difference-aware retrieval architecture can be paired with multimodal action distributions.

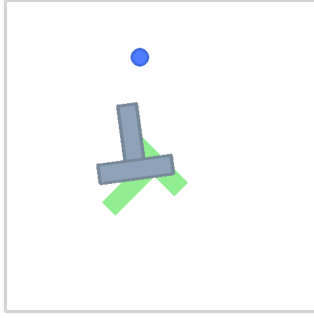


Figure 10: **Push-T Environment.** The goal is to control the blue circle to push the T-shaped block.

Method	Score
BC	48 ± 8
DARP	70 ± 8

Table 4: **Push-T Results.** Averaged over 100 trials, DARP outperforms BC.

A.2.3 CAN DARP HANDLE DISCONTINUOUS ENVIRONMENTS WHERE NEARBY STATES MAY REQUIRE OPPOSING ACTIONS?

A central concern for neighbor-based approaches is performance under discontinuities, where nearby states in Euclidean space may require opposing actions. We evaluate this failure mode with a stress test based on a modified version of D4RL’s Umaze environment (Figure 11).

In this deliberately discontinuous environment, DARP achieves a 57% success rate compared with BC’s 25% (Table 5). The result suggests that difference vectors and permutation-invariant aggregation help the model distinguish useful from misleading neighbors when spatial proximity alone does not guarantee action compatibility.

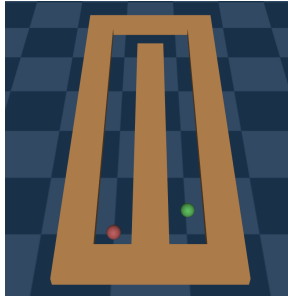


Figure 11: **Long maze environment.** The goal is to move a force-actuated ball from the green start to the red destination.

Method	Succ. (%)
BC	25
DARP	57

Table 5: **Long maze results.** Averaged over 100 trials, DARP outperforms BC in the discontinuous maze stress test.

A.2.4 CAN DARP RECOVER FROM BC ERROR?

To analyze robustness to accumulated error, we roll out a BC agent in settings where it fails and, every k steps, fork the environment state before rolling out DARP from that clone. The results in Fig. 12 show that DARP can recover from many states encountered late in failing BC rollouts. This supports the interpretation that DARP is more robust to moderate accumulated error and can act effectively in slightly OOD states reached by a drifting BC policy.

A.2.5 COMPARISON WITH CCIL

Table 6 compares DARP with CCIL, a baseline that explicitly induces smoothness. Using the scores reported in the CCIL paper, we compare relative improvement over each method’s BC baseline.

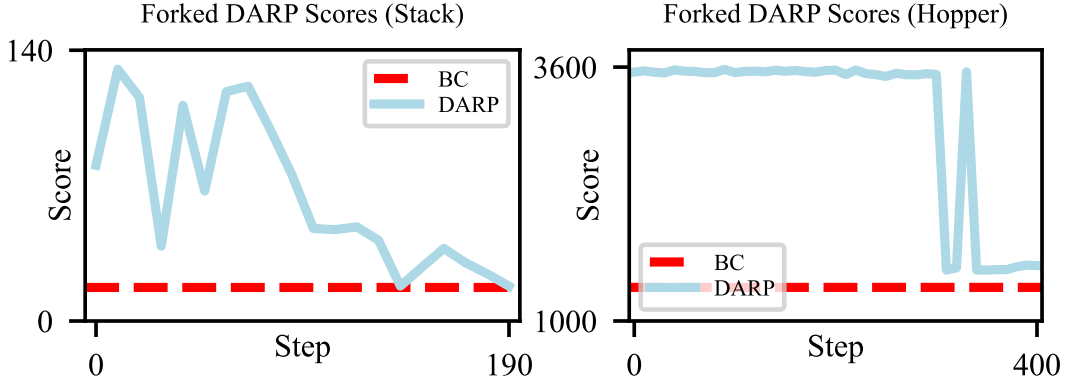


Figure 12: Forked-rollout recovery analysis on Robosuite Stack and MuJoCo Hopper. We roll out a BC agent and create a cloned environment every k steps ($k = 10$ here), then initialize DARP from the cloned state. DARP continues to obtain high scores from many states reached during failing BC rollouts, with performance degrading only about halfway through the Stack rollout and about 80% through the Hopper rollout.

Environment	Method	Improvement over BC (%)
Hopper	CCIL	32.6%
	DARP	53.2%
Walker	CCIL	84.1%
	DARP	84.1%
Ant	CCIL	12.6%
	DARP	84.5%
HalfCheetah	CCIL	5.4%
	DARP	418.7%

Table 6: **Relative improvement over BC compared to CCIL.** Comparing DARP against CCIL on standard MuJoCo benchmarks. DARP achieves higher or equal relative gains across all tasks.

DARP matches CCIL on Walker and achieves larger relative improvements on Hopper, Ant, and HalfCheetah, with the largest difference on HalfCheetah (418.7% vs 5.4% improvement).

A.2.6 DISTANCE METRIC SENSITIVITY

Method	Success Rate (%)
BC	47
DARP (Cosine Similarity)	70
DARP (Euclidean Distance)	75

Table 7: **Distance Metric Sensitivity.** Comparison of success rates using R3M features. DARP with Euclidean distance and cosine similarity perform similarly, beating BC by 28% and 23% respectively. Results are on the Robosuite Stack task.

Although the main experiments use Euclidean distance for retrieval, high-dimensional embeddings such as R3M also motivate alternatives such as cosine similarity. Table 7 shows that DARP with cosine similarity reaches 70% success, compared with 75% for Euclidean distance and 47% for BC. This sensitivity study indicates that performance is not tied exclusively to one distance metric, while also confirming that metric choice affects the final result.

Method	Stack	Threading	Peg Insertion
BC (MLP)	34	12	28
DARP (MLP)	62	38	40
Diffusion	52	44	54
DARP w/ Diffusion	82	72	66

Table 8: **DARP in combination with diffusion policy.** DARP improves both MLP policies and diffusion policies on Robosuite Stack, Threading, and Peg Insertion. Entries are success rates over 50 trials with state-based inputs. The number of demonstrations is smaller than in Table 2, so the BC and DARP values are not directly matched to that table.

A.2.7 DARP IN COMBINATION WITH DIFFUSION POLICY

Diffusion policy (Chi et al., 2024) is a strong model class for imitation learning, particularly in manipulation. Table 8 shows that DARP is complementary to this action model: DARP with a diffusion backbone improves over DARP with an MLP backbone by 20, 34, and 26 percentage points on Stack, Threading, and Peg Insertion, respectively, and improves over standard BC by 48, 60, and 38 percentage points on the same tasks.

Method	Training (s/epoch)	Inference (s/step)
Diffusion	5.610	0.15700
DARP (MLP)	0.559	0.00437

Table 9: **Computational efficiency of DARP with an MLP backbone and diffusion policy.** Comparison of runtime costs between DARP (MLP) and diffusion policy. In this setup, diffusion policy is $\approx 10\times$ slower in training and $\approx 36\times$ slower during inference.

Table 9 also reports that DARP with an MLP backbone is substantially faster than standard diffusion in this setup, especially at inference time.

A.2.8 DARP TRAINED ON HUMAN DEMONSTRATIONS

Method	Success Rate (%)
BC	45
DARP	60

Table 10: **Results with human demonstrations.** Comparison of success rates when training on human data rather than data collected by RL policies. Results are on the Robosuite Stack task.

The MuJoCo and Robosuite demonstrations used in the main experiments are collected from optimized reinforcement-learning policies. To check whether the gain persists with human data, we train on human demonstrations for Robosuite Stack. In this setting, DARP outperforms standard BC by 15 percentage points (Table 10).

A.2.9 CHOICE OF RETRIEVAL HYPERPARAMETERS

Environment	Method	Score (Mean)
Hopper	BC	507.49
	DARP	805.69
Stack	BC	0.12
	DARP	0.31

Table 11: **Performance comparison when validation loss is used to select training epochs and retrieval hyperparameters.** Mean scores on Hopper and Stack environments; DARP maintains a performance gain over BC under validation-based selection.

Figure 9 shows that poor retrieval settings, such as using too few neighbors, can make DARP underperform standard BC. Validation-based selection of training epochs and retrieval hyperparameters nevertheless identifies performant settings in the evaluated Hopper and Stack environments, as shown in Table 11.

A.3 PSEUDOCODE

Algorithm 1 summarizes the training procedure for DARP.

Algorithm 1 Difference-Aware Retrieval Policies

```

1: Input: Expert demonstrations  $\mathcal{D}^* = \{(s_j^*, a_j^*)\}$ , number of neighbors  $k$ 
2: Initialize:  $f$  parameters  $\theta$ 
3: if  $g$  is parametric then
4:   Initialize:  $g$  parameters  $\psi$ 
5: end if
6: // Training Loop
7: while not converged do
8:   Sample batch of query data  $(s_q^*, a_q^*) \sim \mathcal{D}^*$ 
9:   for each query pair  $(s_q^*, a_q^*)$  in batch do
10:    // Find  $k$ -Nearest Neighbors from the entire dataset  $\mathcal{D}^*$ 
11:     $\mathcal{N}_k(s_q^*) \leftarrow \text{argmin-}k_d(s_q^*, s_j^*)$ 
12:    // Compute Neighbor-based Predictions
13:    for each neighbor index  $i \in \mathcal{N}_k(s_q^*)$  do
14:       $a'_i \leftarrow f_\theta(s_i^*, a_i^*, s_i^* - s_q^*)$ 
15:    end for
16:    // Aggregate Predictions
17:    if  $g$  is parametric then
18:       $\hat{a}_q \leftarrow g_\psi(\{a'_i\}_{i \in \mathcal{N}_k(s_q^*)})$ 
19:    else
20:       $\hat{a}_q \leftarrow g(\{a'_i\}_{i \in \mathcal{N}_k(s_q^*)})$ 
21:    end if
22:  end for
23:  // Update Parameters based on the batch loss
24:   $\mathcal{L} \leftarrow \sum_{(s_q^*, a_q^*) \in \text{batch}} \|\hat{a}_q - a_q^*\|^2$ 
25:  // Gradient descent step
26:   $\theta \leftarrow \theta - \alpha \nabla_\theta \mathcal{L}$ 
27:  if  $g$  is parametric then
28:     $\psi \leftarrow \psi - \alpha \nabla_\psi \mathcal{L}$ 
29:  end if
30: end while
31: Output: Trained parameters  $\theta$  and, if applicable,  $\psi$ 

```

A.4 RUNTIME ANALYSIS

Environment	k	Train (s/epoch)	Test (s/step)
Hopper (1,000 datapoints)	BC	0.102	0.00127
	100	0.126	0.00413
	200	0.128	0.00418
	300	0.130	0.00420
	400	0.131	0.00421
	500	0.130	0.00419
Stack (4,200 datapoints)	BC	0.431	0.00139
	100	0.556	0.00437
	250	0.539	0.00442
	500	0.559	0.00437
	750	0.576	0.00433
	1000	0.591	0.00478
	1500	0.682	0.00452
	2000	0.758	0.00451

Table 12: **Runtime comparison across environments.** Training and testing speeds (in seconds) for Behavior Cloning (BC) and varying values of k on Hopper and Stack datasets.

Table 12 reports the training and inference overhead of DARP for different values of k . In these datasets, runtime increases modestly as k grows, and inference remains compatible with real-time robotic control. For example, on Robosuite Stack at $k = 500$, inference takes approximately 0.00437 seconds per step on our hardware, corresponding to a control frequency above 230 Hz.